

A PCA plot showing the relationship between sampling years and life stages. The x-axis is PC1 (47.23%) and the y-axis is PC2 (25.81%). Sampling years are plotted as points, color-coded by life stage: juvenile (purple), ult. landings (red), subadult (green), and adult (orange). Red arrows point from the origin to the centroids of each life stage group. The juvenile group is clustered on the left, the subadult group is clustered on the right, and the adult group is clustered on the right. The ult. landings group is clustered on the left.

